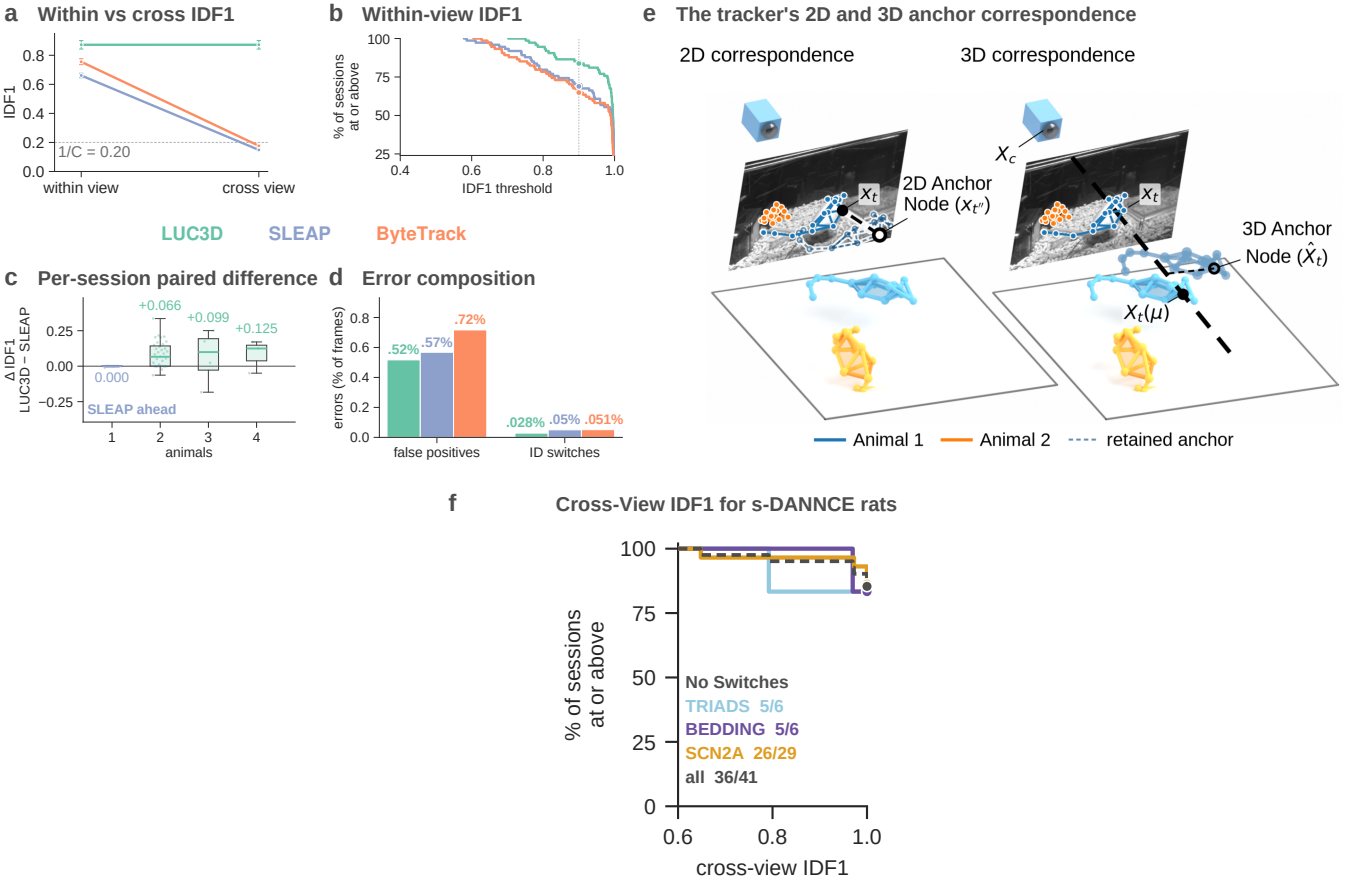

a
Within vs cross IDF1
IDF1
0.8
0.6
0.4
0.2
0.0
1/C = 0.20
within view
cross view
b
Within-view IDF1
% of sessions at or above
100
75
50
25
0.4
0.6
0.8
1.0
IDF1 threshold
LUC3D
SLEAP
ByteTrack
e
The tracker's 2D and 3D anchor correspondence
2D correspondence
3D correspondence
X_c
x_t
2D Anchor Node ($x_{t'}$)
3D Anchor Node ($\hat{X}_t$)
$X_t(\mu)$
Animal 1
Animal 2
retained anchor
c
Per-session paired difference
Δ IDF1 LUC3D − SLEAP
+0.066
+0.099
+0.125
0.000
SLEAP ahead
0.25
0.00
−0.25
1
2
3
4
animals
d
Error composition
errors (% of frames)
0.8
0.6
0.4
0.2
0.0
.52%
.57%
.72%
.028%
.05%
.051%
false positives
ID switches
f
Cross-View IDF1 for s-DANNCE rats
% of sessions at or above
100
75
50
25
0
No Switches
TRIADS 5/6
BEDDING 5/6
SCN2A 26/29
all 36/41
0.6
0.8
1.0
cross-view IDF1